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Crime Data Analysis Using Pig with Hadoop

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Abstract

Big data is the voluminous and complex collection of data that comes from different sources such as sensors, content posted on social media website, sale purchase transaction etc. Such voluminous data becomes tough to process using ancient processing application. There are various tools and techniques in the market for big data analytics. With continually increasing population, crimes and crime rate analyzing related data is a huge issue for governments to make strategic decisions so as to maintain law and order. This is really necessary to keep the citizens of the country safe from crimes. The best place to look up to find room for improvement is the voluminous raw data that is generated on a regular basis from various sources by applying Big Data Analytics (BDA) which helps to analyze certain trends that must be discovered, so that law and order can be maintained properly and there is a sense of safety and well-being among the citizens of the country.

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Keywords: Big data, Hadoop, MapReduce, Sqoop, Flume, Pig

1. Introduction

In recent years, a great proliferation has been witnessed in the amount of data generated. This rate is growing as data is continuously growing. Harnessing such huge amount of data is a tough job and thus it needs some out of the box

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thinking as it cannot be tackled with traditional tools and techniques². This inadequacy led to the birth of the term Big Data and along with it the challenges such as storage, processing, visualization and privacy. Big Data is not just about being big in size. The definition is broadened using five characteristics or “V’s”. These are:

- **Volume:** This characteristic signifies huge voluminous data; it is in orders of terabytes and even petabytes.
- **Velocity:** This characteristic signifies the high velocity with which the data is generated.
- **Variety:** This characteristic refers to the huge variety in the big data.
- **Value:** This characteristic refers to the intrinsic value contained in big data.
- **Veracity:** This characteristic refers to uncertainties in big data such as missing, duplicate and incomplete entries.

Hadoop is an excellent and robust analytics platform for Big Data which can process huge data sets at a really quick speed by providing scalability. It can manage all aspects of Big Data such as volume, velocity and variety by storing and processing the data over a cluster of nodes⁸. Hadoop has two major components in its architecture, MapReduce and the Hadoop Distributed File System (HDFS). With the introduction of YARN (Yet Another Resource Negotiator) in later releases, Hadoop was integrated with a number of wonderful components which can be used for storing, processing and analyzing data a lot more efficiently, thus aiding in exploration of data for undiscovered facts at a smooth pace. Some of these components that work on top of Hadoop are Pig, Flume, Sqoop, Hbase and Oozie.

2. Problem Formulation

In today’s world, every establishment is facing ever growing challenges which need to be coped up quickly and efficiently. With continually increasing population, crimes and crime rate analyzing related data is a huge issue for governments to make strategic decisions so as to maintain law and order. This is really necessary to keep the citizens of the country safe from crimes. The best place to look up to find room for improvement is the voluminous raw data that is generated on a regular basis from various sources by applying Big Data Analytics (BDA). BDA refers to the tools and practices that can be used for transforming the raw data into meaningful and crucial information which helps in forming a decision support system for the judiciary and legislature to take steps towards keeping crimes in check. With the ever increasing population and crime rates, certain trends must be discovered, studied and discussed to take well informed decisions so that law and order can be maintained properly and there is a sense of safety and well-being among the citizens of the country

3. Proposed Framework

3.1. Loading

Apache Sqoop has been designed to move voluminous structured data. It specializes in connecting with RDBMS and moves the structured data into the HDFS. Sqoop has a generic JDBC connector which helps to move data from all kinds of RDBMS’s along with all famous SQL services such as MySQL and oracle’s SQL+. **Apache Flume** is another project which moves data into the HDFS. It is a distributed and reliable service for aggregating, moving and storing unstructured data.

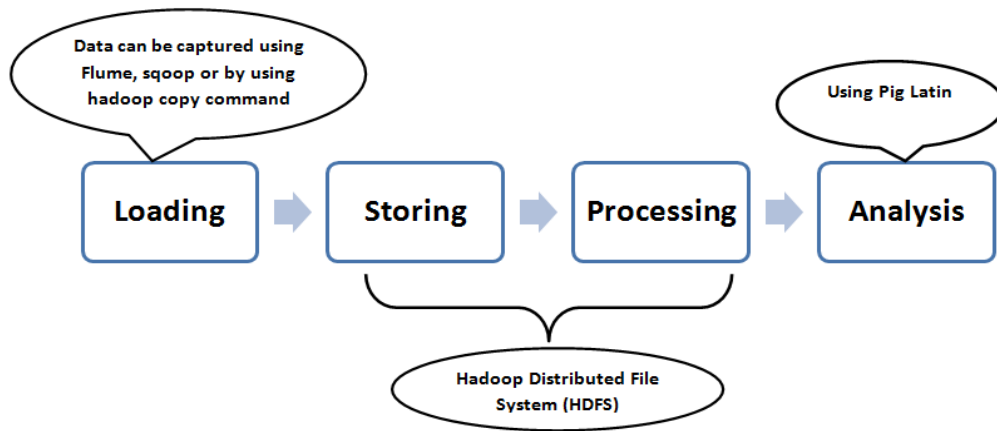


Fig.1 Analysis Framework

3.2. Storing and Processing

The Hadoop distributed file system (HDFS) is a highly scalable and reliable in nature. HDFS is able to store file which is big in size across multiple nodes in a cluster. It is reliable as it replicates the data across the nodes, and hence theoretically does not require RAID (Redundant Array of Integrated Devices) storage. The storage, access and modification jobs are done by two different tasks, the Job tracker (master) and the Task tracker (slave). The Job tracker schedules MapReduce jobs to the Task trackers which know the location of the data. The intercommunication among tasks and nodes is done via periodic messages called heartbeats. One of the ways of achieving this is MapReduce, it is a programming paradigm which can do parallel processing on the nodes in a cluster. It takes input and gives the output in form of key-value pairs⁴. The Map phase processes each record sequentially and independently on every node and generates intermediate key-value pairs. The Mapper processes each record sequentially and independently in parallel and generates intermediate key-value pairs.

$$\text{Map}(k1, v1) \rightarrow \text{list}(k2, v2)$$

The Reduce phase takes the output of the Map phase. It processes and merges all the intermediate values to give the final output, again in form of key-value pairs.

$$\text{Reduce}(k2, \text{list}(v2)) \rightarrow \text{list}(k3, v3)$$

The output gets sorted after each phase, thus providing the user with the aggregated output from all nodes in an orderly fashion.

3.3. Analysis

After collecting the data, it has to be processed so that meaningful information can be extracted out of it which can serve as decision support system. Therefore the analysts need to come up with a good technique for the same. However, Writing MapReduce requires basic knowledge of Java along with sound programming skills. Even after writing the code, which is in itself a labor intensive task, with additional time is required for the review of code and its quality assessment. But now, analysts have additional options of using the Pig Latin. Pig Latin is the scripting language to construct MapReduce programs for an Apache project which runs on Hadoop. The benefit of using this is that fewer lines of code has to be written which reduces overall development and testing time. These scripts take just about 5% time compared to writing MR programs but 50% more time in execution. Although Pig scripts are

50% slower in execution compared to MR programs, they still are very effective in increasing productivity of data engineers and analysts by saving lots of time in writing phase.

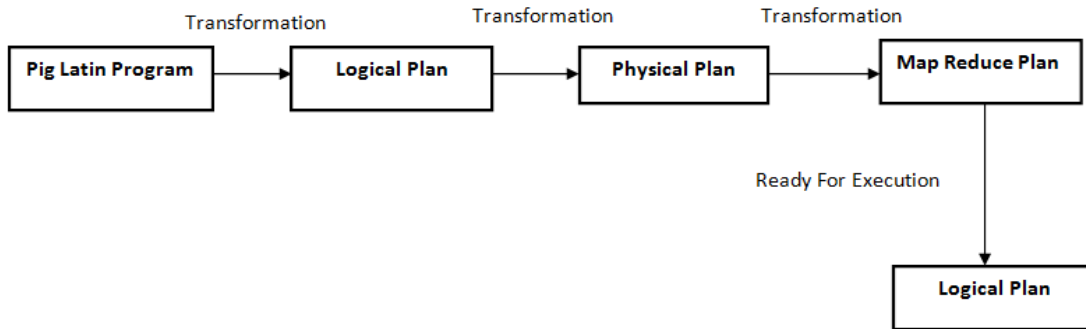


Fig. 2 Pig Latin Workflow

Pig is scripting language. The script consists of data flow instructions which are first converted into logical plan in which it allows to optimize the process and guarantees that the users' directives are followed. In the physical plan pig allows for optimization of joins, grouping and parallelism of tasks. Also, different operators can be having different factor of parallelization to cope with the varying volumes of data flow. The script then converted to MapReduce instructions by its framework and used to process data.

4. Research Questions

Some of the problem statements along which the analysis has been done in this paper:

1. Total Number of crimes from the year 2000-2014

With the ever increasing population and crime rates, certain trends must be discovered, so that law and order can be maintained properly and there is a sense of safety and well-being among the citizens of the country.

2. Total Number of crimes occurring in each state

If the number of complaints from a particular state is found to be very high, extra security must be provided to the residents there by increasing police presence, quick redressal of complaints and strict vigilance.

3. Total Number of Crime by Type

It includes crime in which the objective is violent for example murder or in which violence means to an end for example robbery.

4. Total Number of crimes against women

Crimes against women are becoming an increasingly worrying and disturbing problem for the government. The number of such crimes must be found, especially the ones against young women (age between 18-30 years)

Hadoop Cluster Mode: Pseudo-Distributed hadoop cluster mode is used that is hadoop daemons run on the local machine.

Data Set: The data was in .csv Format, that is each line represents data record and each record has one or more field separated by commas and was containing symbols like parenthesis, hyphens etc.

Data Set Description: The data set includes the following fields:

- The date of the crime along with year
- The age and gender of the accused
- The type of offence committed

- The age and gender of the victim. This field is divided into a number of age ranges (below 18, 18-30, 30-45, 45-60, above 60)

Tools and Technologies Used:

1. Hadoop
2. Pig

5. Experimental Findings

5.1. Total Number of crimes from the year 2000-2014

Input: Crime data set

Output: Total number of crime from the year 2000-2014

Algorithm used in Grunt Shell.

1. Enter into the Grunt shell using command : **Pig**
2. X = Load the data set using Pig Storage;
3. Y = For each X generate the crime by year;
4. Z = Group by year;
5. Data = For each Z generate group, SUM(X.year);
6. Store output.

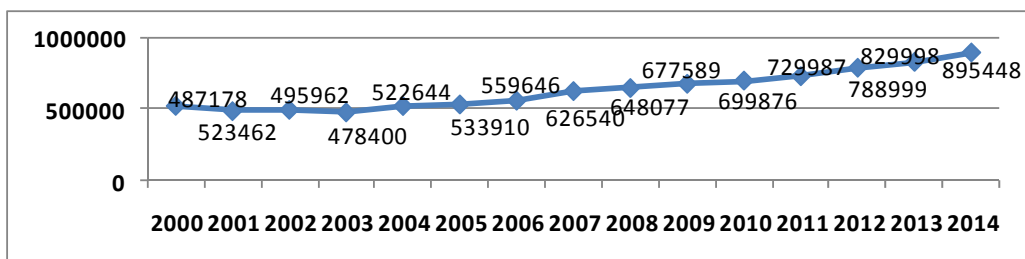


Fig. 3 Total Number of crimes from the year 2000-2014

From the above graph it can be concluded that there is a considerable amount of growth in crime from the year 2000 to 2014. With the ever crime rates, certain trends must be discovered, studied and discussed to take well informed decisions so that law and order can be maintained properly and there is a sense of safety and well-being among the citizens of the country

5.2. Total Number of crimes occurring in each state

Input: Crime data set

Output: Total number of crime occurring in each state

Algorithm used in Grunt Shell.

1. Enter into the Grunt shell using command : **Pig**
2. X = Load the data set using Pig Storage;
3. Y = For each X generate the crime year by state;

4. $Z = \text{Group by state};$
5. $\text{Data} = \text{For each } Z \text{ generate group, SUM}(X.\text{year});$
6. Store output;

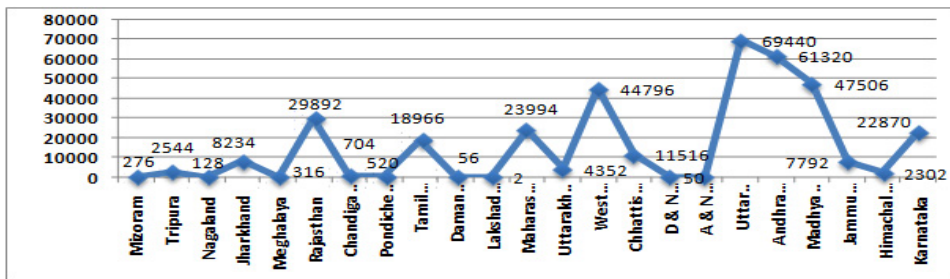


Fig. 4 Total Number of Crimes Occurring In Each State

From the above graph it is clear that if the number of complaints from a particular state is found to be very high, extra security must be provided to the residents there by increasing police presence, quick redressal of complaints and strict vigilance.

5.3. Total Number of Crime by Type

Input: Crime data set

Output: Total number of crime by type.

Algorithm used in Grunt Shell.

1. Enter into the Grunt shell using command : **Pig**
2. $X = \text{Load the data set using Pig Storage};$
3. $Y = \text{For each } X \text{ generate the crime by crime head};$
4. $Z = \text{Group by crime head};$
5. $\text{Data} = \text{For each } Z \text{ generate group, SUM}(X.\text{year});$
6. Store output;

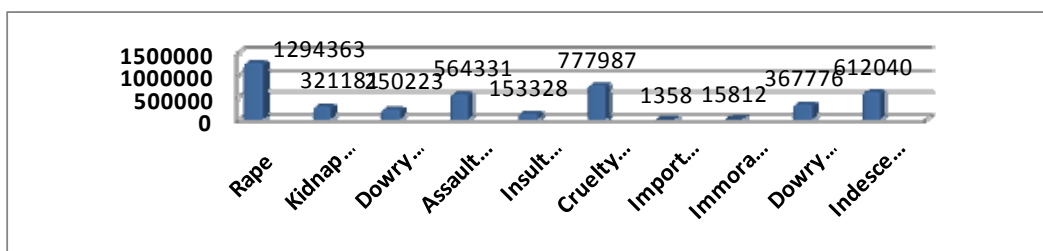


Fig. 5 Total Number of crime by type

Above graph includes crime in which the objective is violent with the ever increasing population and crime rates, certain trends must be discovered, so that law and order can be maintained properly and there is a sense of safety and well-being among the citizens of the country.

5.4. Total Number of crimes against women

Input: Crime data set

Output: Total number of crime against women.

Algorithm Used In Grunt Shell:

1. Enter into the Grunt shell using command : **Pig**
2. **X** = Load the data set using Pig Storage;
3. **Y** = For each **X** generate the crime age by state and find number of females who were victims in different crimes.
4. **Z** = Group by state;
5. **Data** = For each **Z** Generate group, SUM(**Y.age**);
6. Store output;

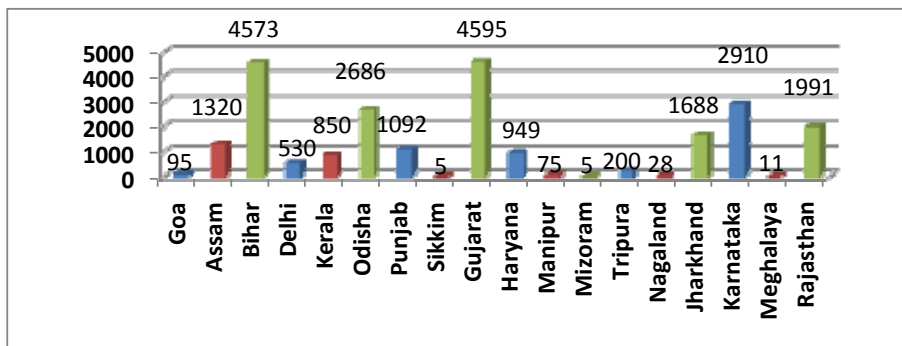


Fig. 6 Total Number of Crime against Women

Crimes against women are becoming an increasingly worrying and disturbing problem for the government. Extra security must be provided so that law and order can be maintained properly and there is a sense of safety and well-being among the citizens of the country

6. Scope

The various limitations of this analysis:

- There could be more techniques and area of applications, therefore as authors we cannot claim for thorough research
- Pseudo-Distributed hadoop cluster mode is used that is hadoop daemons run on the local machine.
- The extent of the search was confined to few publishers which is the prime limitation of the search.
- Non English publications were excluded, we believe research regarding the application has been discussed and published in other languages.

7. Conclusion and Future Work

Big Data Analytics refers to the tools and practices that can be used for transforming this raw data into meaningful and crucial information which helps in forming a decision support system for the judiciary and legislature to take steps towards keeping crimes in check. With the ever increasing population and crime rates, certain trends must be discovered, studied and discussed to take well informed decisions so that law and order can be maintained properly. If the number of complaints from a particular state is found to be very high, extra security must be provided to the residents there by increasing police presence, quick redressal of complaints and strict vigilance. Crimes against women are becoming an increasingly worrying and disturbing problem for the government. The number of such crimes must be found, especially the ones against young women (age between 18-30 years). Extra security must be

provided so that law and order can be maintained properly and there is a sense of safety and well-being among the citizens of the country.

7.1. Future work:

- This analysis can be further carried out on Fully Distributed cluster mode that is hadoop daemons run on a cluster of machine
- Similar analysis can be further carried out in different sector.

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